THE COMPLETE GUIDE · 2026

AI/ML ROADMAP 2026

From zero to AI Agents — the complete path every engineer needs in 2026.

01 Mathematics
02 Python
03 ML Basics
04 Data Skills
05 Deep Learning
06 Generative AI
07 NLP & LLMs
08 MLOps
09 AI Agents

2026

STAGE 01 · FOUNDATION

01

Mathematics

You don't need a PhD. You need these 4 areas.

CORE AREAS

- **Linear Algebra —** Vectors, matrices, dot products, eigenvalues. Every ML operation is matrix math.
- **Calculus —** Derivatives, chain rule, gradients. Backpropagation is just chain rule applied repeatedly.
- **Statistics —** Distributions, hypothesis testing, Bayes. Data without statistics is just noise.
- **Probability —** Conditional probability, expectations, sampling. The language of uncertainty.

→ Start here: 3Blue1Brown (YouTube) · Khan Academy · Mathematics for ML book (free PDF online).

STAGE 02 · FOUNDATION

02 Python for Data

Learn Python like a data person, not a software engineer.

CORE LIBRARIES

- **NumPy —** Arrays, broadcasting, math operations. Backbone of all numerical computing.
- **pandas —** DataFrames, cleaning, merging, groupby. Your daily driver for every data task.
- **matplotlib —** Visualisation. If you can't plot it, you don't truly understand it yet.
- **Jupyter —** Interactive notebooks. Think, run, inspect, iterate — all in one environment.

→ Graduation test: Load a CSV, clean it, plot it, export it without Googling. Then you're ready.

STAGE 03 · CORE

Machine Learning

Your first real models. sklearn is your best friend here.

03 SUPERVISED LEARNING

- **Regression & Classification —** Linear, Logistic, Decision Trees, SVM, KNN. Understand when to use each.
- **Ensemble Methods —** Random Forest and XGBoost. These win competitions and ship to production.

MUST KNOW

- **Train / Val / Test Split —** The single most important habit in ML. Never skip this step.
- **Evaluation Metrics —** AUC, F1, RMSE. Always pick the metric that matches your business goal.

STAGE 04 · CRITICAL

Data Skills

80% of ML is this. Most tutorials skip it entirely.

04

DATA CLEANING

- **Missing Values —** Detect, drop, impute. Median for numerical, mode for categorical columns.
- **Outlier Handling —** IQR and Z-Score methods. Cap real extremes, remove clear data errors.
- **Feature Scaling —** MinMaxScaler vs StandardScaler. Wrong choice breaks distance models silently.
- **Encoding —** One-Hot for nominal, Ordinal for ordered categories. Label on nominal = silent bugs.

FEATURE ENGINEERING

- **EDA —** Distributions, correlations, Seaborn. Ask questions before you model anything.
- **New Features —** Interaction terms, log transforms. Domain knowledge wins competitions here.

STAGE 05 · ADVANCED

Deep Learning

Neural networks. Where the real magic actually happens.

05

FOUNDATIONS

- **Network Basics —** Perceptrons, layers, activation functions. Build one from scratch first.
- **Backpropagation —** Chain rule in action. Understand this deeply before touching PyTorch.
- **Optimisers —** Adam, SGD, learning rate schedules. Adam is your safe default starting point.

ARCHITECTURES

- **CNNs —** Convolutional networks for images — classification, detection, segmentation.
- **Transformers —** Attention is all you need. The architecture powering everything modern in 2026.

FRAMEWORK

- **PyTorch —** Recommended for 2026. Research-first, flexible, industry is steadily moving here.

STAGE 06 · FRONTIER

Generative AI

From discriminating to creating. The paradigm shift.

CORE MODELS

- **Diffusion Models —** DALL·E, Stable Diffusion, Flux. The dominant approach for generation in 2026.
- **GANs & VAEs —** Adversarial training, latent spaces. Learn these before diffusion makes full sense.
- **Multimodal LLMs —** GPT-4o, Gemini, Claude. Text, image and audio handled in one single model.

06

KEY SKILLS

- **Fine-tuning —** LoRA, DreamBooth. Adapt base models to your specific domain cheaply and fast.
- **Prompt Engineering —** Chain-of-thought, few-shot, structured outputs. Systematic, not guesswork.

GEN

STAGE 07 · HOT

NLP & LLMs

Language is the interface. Learn to work with it properly.

FUNDAMENTALS

- **Tokenisation & Embeddings —** How text becomes numbers. Word2Vec, sentence transformers, vector spaces.
- **Transformer Architecture —** Attention heads, positional encoding, encoder-decoder. Read the original paper.

MODERN LLMS

07 • **Working with LLMs —** GPT, Claude, LLaMA. API usage, context management, structured outputs.

- **RAG —** Retrieval Augmented Generation. Ground LLMs in your own data. The #1 enterprise AI pattern.
- **Fine-tuning —** QLoRA for efficient open-source model adaptation on domain-specific data.

STAGE 08 · PRODUCTION

MLOps & Deployment

A model no one can use is not a model.

SERVING

- **FastAPI + Docker —** Model serving APIs, fully containerised. REST and async inference patterns.
- **Inference Optimisation —** Quantisation, batching, response caching. Speed matters with real users.

PIPELINES & MONITORING

- **MLflow / DVC —** Experiment tracking and data versioning. Always know exactly what you ran.
- **CI/CD for ML —** Automated testing, model registry, deployment pipelines. Treat models like software.
- **Drift Detection —** Models decay silently in production. Monitor your data distributions always.

STAGE 09 · FRONTIER

AI Agents

Build systems that think, plan and act. The 2026 frontier.

CORE CONCEPTS

- **ReAct Pattern —** Reason + Act loop. How agents decide what to do next given a goal and tools.
- **Tool Use —** Function calling, web search, code execution. Give your LLMs actual hands.
- **Memory —** Short-term context, long-term vector store, episodic. Agents that actually remember.

FRAMEWORKS

- **LangGraph —** Stateful multi-step agent workflows. Production-grade choice for agents in 2026.
- **CrewAI —** Multi-agent systems. Specialised agents collaborating on complex multi-step tasks.
- **MCP Protocol —** Anthropic's tool integration standard. The emerging industry convention.

YOUR MOVE.

You don't need all 9 stages to get hired. Pick one. Go deep. Ship something real.

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Save this. Share it.

Which stage are you at? Comment below.

GO.